

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: Differential Equation & Linear Algebra

Time: 1.5 Hours

Semester / Programme: First Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) Find the integrating factor of the first-order linear differential equation $x \frac{dy}{dx} + yx = 1$. [1]

$$I.F. = e^{\int P(x)dx}$$

Solution & Key Concept:

- Rewrite the given equation in standard form $(dy/dx) + P(x)y = Q(x)$ by dividing throughout by x : $(dy/dx) + (x)y = 1/x$.
- Identify $P(x) = x$.
- Calculate the integrating factor: $I.F. = e^{\int x dx} = e^{|x|} = x$.

Answer: $\sec x$

Marking Rubric: [0.5 Mark] for reducing to standard form and identifying $P(x)$, [0.5 Mark] for correct evaluating of integrating factor.

1(b) Determine the Wronskian $W(y_1, y_2)$ of the two solutions $y_1 = e^{2x}$ and $y_2 = e^{-3x}$ of a second-order linear differential equation. [1]

$$\text{Formula: } W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Solution & Key Concept:

- Find the derivatives: $y_1' = 2e^{2x}$ and $y_2' = -3e^{-3x}$.
- Compute the determinant: $W = \begin{vmatrix} e^{2x} & e^{-3x} \\ 2e^{2x} & -3e^{-3x} \end{vmatrix} = (e^{2x})(-3e^{-3x}) - (e^{-3x})(2e^{2x})$.
- Simplify: $W = -3e^{-x} - 2e^{-x} = -5e^{-x}$.

Answer: $-5e^{-x}$

Marking Rubric: [0.5 Mark] for setting up Wronskian determinant with derivatives, [0.5 Mark] for final simplification.

1(c) Evaluate the improper integral $\int_1^\infty (1/x^2) dx$. [1]

$$\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$$

Solution & Key Concept:

- Express as a limit: $\int_1^\infty x^{-2} dx$.
- Integrate: $\left[-(1/x) \right]_1^t = -(1/t) - (-1) = 1 - (1/t)$.
- Evaluate the limit as $t \rightarrow \infty$: $1 - 0 = 1$.

Answer: 1

Marking Rubric: [0.5 Mark] for limit formulation and integration, [0.5 Mark] for correct evaluation.

1(d) State the relationship between the Gamma function ($\Gamma(n)$) and the Beta function $B(m, n)$ for positive real numbers m and n . [1]

$$B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

Solution & Key Concept:

- Recall the definition of Beta function in terms of Gamma function.
- Write down the standard algebraic relation.

Answer: $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

Marking Rubric: [1 Mark] for correct formula statement.

1(e) Find the general form of the complementary function (C.F.) for the second-order differential equation $(D^2 - 4D + 4)y = 0$. [1]

$$m^2 - 4m + 4 = 0$$

Solution & Key Concept:

- Write the auxiliary equation: $m^2 - 4m + 4 = 0$.
- Solve for roots: $(m - 2)^2 = 0$ $m = 2, 2$ (repeated real roots).
- Write the complementary function for repeated roots: $y_c = (C_1 + C_2 x)e^{2x}$.

Answer: $(C_1 + C_2 x)e^{2x}$

Marking Rubric: [0.5 Mark] for auxiliary equation and roots, [0.5 Mark] for correct C.F. structure with repeated roots.

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Solve the exact differential equation: $(2x + y^3)dx + (3xy^2 + 4)dy = 0$. [2.5]

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution for (a):

- Identify $M = 2x + y^3$ and $N = 3xy^2 + 4$.
- Check exactness: $(M / y) = 3y^2$ and $(N / x) = 3y^2$. Since $(M / y) = (N / x)$, the equation is exact.
- Integrate M with respect to x treating y as constant: $\int (2x + y^3) dx = x^2 + xy^3$.
- Integrate terms of N free of x with respect to y : $\int 4 dy = 4y$.
- Combine to form the general solution: $x^2 + xy^3 + 4y = C$.

Result / Conclusion: $x^2 + xy^3 + 4y = C$

Marking Rubric: [1 Mark] for checking exactness, [1 Mark] for integration steps, [0.5 Mark] for general solution.

(b) Find the orthogonal trajectories of the family of curves given by $y = Cx^2$, where C is an arbitrary constant. [2.5]

$$\frac{dy}{dx} \rightarrow -\frac{dx}{dy}$$

Solution for (b):

- Differentiate the given equation implicitly with respect to x : $(dy / dx) = 2Cx$.
- Eliminate the arbitrary constant C using $C = (y / x^2)$: $(dy / dx) = 2 \frac{y}{x}$.
- Replace (dy / dx) with $-(dx / dy)$ to get the differential equation for orthogonal trajectories: $-(dx / dy) = (2y / x) \times dx = -2y dy$.
- Integrate both sides: $\int x dx = -2 \int y dy$ $(x^2 / 2) = -y^2 + K_1$ $x^2 + 2y^2 = C'$.

Result / Conclusion: $x^2 + 2y^2 = C'$

Marking Rubric: [1 Mark] for finding slope and eliminating parameter, [1 Mark] for substituting orthogonal slope condition, [0.5 Mark] for final integrated curve.

- Q.3. (a)** Solve the Bernoulli differential equation: $(dy/dx) + (y/x) = x^2 y^2$. [2.5]
- $v = y^{1-n}$

Solution for (a):

- Here $n = 2$. Divide the entire equation by y^2 : $y^{-2}(dy/dx) + (1/x)y^{-1} = x^2$.
- Substitute $v = y^{-1}$, so $(dv/dx) = -y^{-2}(dy/dx)$, which gives $y^{-2}(dy/dx) = -(dv/dx)$.
- Substitute into the transformed equation: $-(dv/dx) + (1/x)v = x^2$ $(dv/dx) - (1/x)v = -x^2$.
- This is linear in v . Integrating factor $I.F. = e^{\int \{-1/x\} dx} = e^{-\ln x} = (1/x)$.
- Multiply and solve: $v \leq \int \{(1/x)\} = \int \{-x^2\} \leq \int \{(1/x)\} dx = \int -x dx = -(x^2/2) + C$.
- Substitute back $v = (1/y)$: $(1/xy) = -(x^2/2) + C$.

Result / Conclusion: $\frac{1}{xy} + \frac{x^2}{2} = C$

Marking Rubric: [1 Mark] for substitution and linear form reduction, [1 Mark] for finding integrating factor and solving linear ODE, [0.5 Mark] for back-substitution.

- (b)** A thermometer reading 70°F is placed in an oven preheated to a constant temperature of 300°F . [2.5]
Through Newton's Law of Cooling/Heating, if the thermometer reads 150°F after 2 minutes, determine the temperature at $t = 5$ minutes.

$$\frac{dT}{dt} = -k(T - T_m)$$

Solution for (b):

- Set up the differential equation: $(dT/dt) = k(300 - T)$ with $T(0) = 70$.
- Separate variables and integrate: $(dT/300 - T) = k dt$ $-(300 - T) = kt + C_1$ $300 - T = C e^{-kt}$.
- Use initial condition $T(0) = 70$: $300 - 70 = C e^0$ $C = 230$. So $T(t) = 300 - 230e^{-kt}$.
- Use condition $T(2) = 150$: $150 = 300 - 230e^{-2k}$ $150 = 230e^{-2k}$ $e^{-2k} = (15/23)$.
- Find $T(5)$: $T(5) = 300 - 230(e^{-2k})^{5/2} = 300 - 230 \leq \int \{(15/23)\}^{2.5}$.
- Calculate numerical value: $T(5) \approx 300 - 230(0.355) \approx 300 - 81.65 = 218.35^\circ\text{F}$.

Result / Conclusion: $T(5) \approx 218.35^\circ\text{F}$

Marking Rubric: [1 Mark] for differential equation setup and finding constant C , [1 Mark] for determining decay constant factor via $t=2$ condition, [0.5 Mark] for final evaluation at $t=5$.

- Q.4. (a)** Solve the Euler-Cauchy equation: $x^2 y'' - 3xy' + 4y = 0$. [2.5]
 $m(m-1) + am + b = 0$

Solution for (a):

- Substitute $x = e^z$ (i.e., $z = \ln x$), where $x D = \theta$ and $x^2 D^2 = \theta(\theta - 1)$ with $\theta = (d/dz)$.
- Transform the differential equation: $[\theta(\theta - 1) - 3\theta + 4]y = 0 \Rightarrow (\theta^2 - 4\theta + 4)y = 0$.
- Write auxiliary equation: $m^2 - 4m + 4 = 0 \Rightarrow (m - 2)^2 = 0 \Rightarrow m = 2, 2$.
- Write solution in terms of z : $y(z) = (C_1 + C_2 z)e^{2z}$.
- Convert back to variable x : $y(x) = (C_1 + C_2 \ln x)x^2$.

Result / Conclusion: $y(x) = (C_1 + C_2 \ln x)x^2$

Marking Rubric: [1 Mark] for substitution and auxiliary equation, [1 Mark] for general solution in z , [0.5 Mark] for returning to variable x .

- (b)** Find the particular integral (P.I.) of the differential equation $(D^2 + 4)y = (2x)$ using operator methods. [2.5]
 $P.I. = \frac{1}{f(D)}Q(x)$

Solution for (b):

- Write the expression for P.I.: $P.I. = \frac{1}{D^2 + 4}(2x)$.
- Since $Q(x) = (ax)$ with $a = 2$, replace D^2 with $-a^2 = -(2^2) = -4$.
- Notice failure of method because denominator becomes $-4 + 4 = 0$: $f(-a^2) = 0$.
- Apply the failure formula: $P.I. = x \frac{1}{\{d\} \{DD\} (D^2 + 4)} (2x) = x \frac{1}{2D} (2x)$.
- Evaluate integration: $\frac{1}{D} (2x) = \int 2x dx = (2x^2)/2$.
- Combine terms: $P.I. = x \cdot \frac{1}{2} \cdot (2x) = x^2$.

Result / Conclusion: $\frac{x^2}{4} \cos(2x)$

Marking Rubric: [1 Mark] for identifying denominator zero failure condition, [1 Mark] for applying rule of multiplying by x and differentiating denominator, [0.5 Mark] for final integration result.

- Q.5. (a)** Solve the nonhomogeneous differential equation $y'' - 2y' + y = (e^x / x)$ using the method of variation of parameters. [2.5]

$$P.I. = -y_1 \int \frac{y_2 Q}{W} dx + y_2 \int \frac{y_1 Q}{W} dx$$

Solution for (a):

- Find complementary function: $m^2 - 2m + 1 = 0$ $m = 1, 1$. So $y_1 = e^x$, $y_2 = x e^x$, and $C.F. = C_1 e^x + C_2 x e^x$.
- Compute Wronskian $W(y_1, y_2)$: $W = \begin{vmatrix} e^x & x e^x \\ e^x & (1+x)e^x \end{vmatrix} = e^{2x}(1+x) - x e^{2x} = e^{2x}$.
- Identify $Q(x) = (e^x / x)$.
- Calculate coefficients u_1 and u_2 : $u_1' = -(y_2 Q / W) = -(x e^x \cdot (e^x / x) / e^{2x}) = -1$ $u_1 = -x$.
- $u_2' = (y_1 Q / W) = (e^x \cdot (e^x / x) / e^{2x}) = (1 / x)$ $u_2 = \ln x$.
- Construct particular integral: $P.I. = u_1 y_1 + u_2 y_2 = (-x)(e^x) + (\ln x)(x e^x) = -x e^x + x e^x \ln x$.

Result / Conclusion: $y = C_1 e^x + C_2 x e^x + x e^x (\ln x - 1)$

Marking Rubric: [1 Mark] for C.F. and Wronskian, [1 Mark] for integration of variation parameter formulas, [0.5 Mark] for combining into general solution.

- (b)** Test the convergence of the improper integral $\int_0^1 (1 / x^{1/2}) dx$. [2.5]

$$\int_0^a \frac{1}{x^p} dx \text{ converges if } p < 1$$

Solution for (b):

- Identify that the integrand has an infinite discontinuity at the lower limit $x = 0$.
- Express as a limit: $\lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} dx$.
- Evaluate the antiderivative: $\int_t^1 x^{-1/2} dx = \left[2x^{1/2} \right]_t^1 = 2(1)^{1/2} - 2(t)^{1/2} = 2 - 2\sqrt{t}$.
- Take the limit as $t \rightarrow 0^+$: $\lim_{t \rightarrow 0^+} (2 - 2\sqrt{t}) = 2 - 0 = 2$.
- Since the limit exists and is finite, the improper integral converges.

Result / Conclusion: Converges to 2

Marking Rubric: [1 Mark] for limit formulation of improper integral with singularity, [1 Mark] for correct integration, [0.5 Mark] for limit evaluation and convergence conclusion.

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